Name
Date
Course

Solving Systems of Three Equations

Substitution

Solve.

① $\begin{cases} 5x-2y+z=-4 \\ 3x+y+2z=-3 \\ -2y=z \end{cases}$

$5x-2y-2y=-4$

$5x-4y=-4$

$3x+y+2(-2y)=-3$

$3x+y-4y=-3$

$3x-3y=-3$

$x-y=-1$

$\begin{cases} 5x-4y=-4 \\ x-y=-1 \end{cases}$

$\begin{cases} 5x-4y=-4 \\ -5x+5y=5 \end{cases}$

$\boxed{y=1}$

$x-1=-1$

$\boxed{x=0}$

$z=-2(1)$

$\boxed{z=-2}$

② $\begin{cases} x-4y+2z=5 \\ y=-2-x \\ z+2x=-1 \end{cases}$

$x-4(-2-x)+2z=5$

$x+8+4x+2z=5$

$5x+8+2z=5$

$5x+2z=-3$

$\begin{cases} 2x+z=-1 \\ 5x+2z=-3 \end{cases}$

$\begin{cases} 4x+2z=-2 \\ 5x+2z=-3 \end{cases}$

$-x=1$

$\boxed{x=-1}$

$y=-2-(-1)$

$y=-2+1$

$\boxed{y=-1}$

$2(-1)+z=-1$

$-2+z=-1$

$\boxed{z=1}$

③ $\begin{cases} x = y-3+z \\ x-2y+z=-1 \\ 2x-y+3z=6 \end{cases}$

$y-3+z-2y+z=-1$

$-y+2z=2$

$2(y-3+z)-y+3z=6$

$2y-6+2z-y+3z=6$

$y+5z=12$

$\begin{cases} -y+2z=2 \\ y+5z=12 \end{cases}$

$7z=14$

$\boxed{z=2}$

$y+5(2)=12$

$y+10=12$

$\boxed{y=2}$

$x=2-3+2$

$\boxed{x=1}$

④ $\begin{cases} z+2=x+y \\ x=y+z \\ y=z+3 \end{cases}$

$z+2=y+z+y$

$z+2=2y+z$

$2=2y$

$\boxed{1=y}$

$1=z+3$

$\boxed{-2=z}$

$x=1+(-2)$

$\boxed{x=-1}$

5. $\begin{cases} 4y = x \\ -2x + 3y - z = 1 \\ x + y + z = 2 \end{cases}$

$-2(4y) + 3y - z = 1$

$-8y + 3y - z = 1$

$-5y - z = 1$

$4y + y + z = 2$

$5y + z = 2$

$\begin{cases} -5y - z = 1 \\ 5y + z = 2 \end{cases}$

$0 = 3$

$\boxed{\text{No Solutions}}$

6. $\begin{cases} 10x - 3y + 4z = -22 \\ y = 5x \\ 2x - y - 3z = 3 \end{cases}$

$10x - 3(5x) + 4z = -22$

$10x - 15x + 4z = -22$

$-5x + 4z = -22$

$2x - 5x - 3z = 3$

$-3x - 3z = 3$

$x + z = -1$

$\begin{cases} -5x + 4z = -22 \\ x + z = -1 \end{cases}$

$\begin{cases} -5x + 4z = -22 \\ 4x + 4z = -4 \end{cases}$

$-9x = -18$

$\boxed{x = 2}$

$y = 5(2)$

$\boxed{y = 10}$

$2 + z = -1$

$\boxed{z = -3}$

⑦ $\begin{cases} 3x+7y-9z=-22 \\ -4x+z=-2 \\ x=2+y \end{cases}$

$$3(2+y)+7y-9z=-22$$
$$6+3y+7y-9z=-22$$
$$10y-9z=-28$$

$$-4(2+y)+z=-2$$
$$-8-4y+z=-2$$
$$-4y+z=6$$

$\begin{cases} 10y-9z=-28 \\ -4y+z=6 \end{cases}$

$\begin{cases} 10y-9z=-28 \\ -36y+9z=54 \end{cases}$

$$-26y=26$$
$$\boxed{y=-1}$$

$$x=2+(-1)$$
$$\boxed{x=1}$$

$$-4(-1)+z=6$$
$$4+z=6$$
$$\boxed{z=2}$$

⑧ $\begin{cases} 2y+1=z \\ -x+3y-4z=-5 \\ x-y+z=2 \end{cases}$

$$-x+3y-4(2y+1)=-5$$
$$-x+3y-8y-4=-5$$
$$-x-5y=-1$$

$$x-y+2y+1=2$$
$$x+y=1$$

$\begin{cases} -x-5y=-1 \\ x+y=1 \end{cases}$

$$-4y=0$$
$$\boxed{y=0}$$

$$x+0=1$$
$$\boxed{x=1}$$

$$2(0)+1=z$$
$$\boxed{1=z}$$

⑨ $\begin{cases} x = z \\ 5x - 4 = y \\ 2x - y = z - 4 \end{cases}$

$5z - 4 = y$

$2z - y = z - 4$

$\begin{cases} 5z - 4 = y \\ 2z - y = z - 4 \end{cases}$

$2z - (5z - 4) = z - 4$

$2z - 5z + 4 = z - 4$

$-3z + 4 = z - 4$

$8 = 4z$

$\boxed{2 = z}$

$x = 2$

$5(2) - 4 = y$

$\boxed{6 = y}$

⑩ $\begin{cases} 2x + y - 3z = 0 \\ x = y - z \\ 4x + 2y - 6z = 0 \end{cases}$

$2(y - z) + y - 3z = 0$

$2y - 2z + y - 3z = 0$

$3y - 5z = 0$

$4(y - z) + 2y - 6z = 0$

$4y - 4z + 2y - 6z = 0$

$6y - 10z = 0$

$\begin{cases} 3y - 5z = 0 \\ 6y - 10z = 0 \end{cases}$

$\begin{cases} -6y + 10z = 0 \\ 6y - 10z = 0 \end{cases}$

$0 = 0$

$\boxed{\text{Infinite Solutions}}$

Elimination

(11) $\begin{cases} x+2y+z=-3 & a \\ 3x-y+z=10 & b \\ 5x+4y-z=-3 & c \end{cases}$

$a+c \rightarrow 6x+6y=-6$

$b+c \rightarrow 8x+3y=7$

$\frac{a+c}{6} \rightarrow x+y=-1$

$\begin{cases} 8x+3y=7 \\ x+y=-1 \end{cases}$

$\begin{cases} 8x+3y=7 \\ -3x-3y=3 \end{cases}$

$5x=10$

$\boxed{x=2}$

$2+y=-1$

$\boxed{y=-3}$

$2+2(-3)+z=-3$

$2-6+z=-3$

$\boxed{z=1}$

(12) $\begin{cases} -x+3y+2z=6 & a \\ 2x-y+z=-2 & b \\ -x+2y-4z=-1 & c \end{cases}$

$a-c \rightarrow y+6z=7$

$\begin{cases} -2x+6y+4z=12 & 2a \\ 2x-y+z=-2 & b \end{cases}$

$5y+5z=10$

$y+z=2$

$\begin{cases} y+6z=7 \\ y+z=2 \end{cases}$

$5z=5$

$\boxed{z=1}$

$y+1=2$

$\boxed{y=1}$

$2x-1+1=-2$

$2x=-2$

$\boxed{x=-1}$

(13) $\begin{cases} 5x+2y+10z=7 & a \\ 3x+4y-5z=7 & b \\ 2x+3y+8z=5 & c \end{cases}$

$-3a \begin{cases} -15x-6y-30z=-21 \\ 4x+6y+16z=10 \end{cases} 2c$

$-11x-14z=-11$

$-2a \begin{cases} -10x-4y-20z=-14 \\ 3x+4y-5z=7 \end{cases} b$

$-7x-25z=-7$

$\begin{cases} -7x-25z=-7 \\ -11x-14z=-11 \end{cases}$

$\begin{cases} 77x+275z=77 \\ -77x-98z=-77 \end{cases}$

$177z=0$

$\boxed{z=0}$

$-7x-25(0)=-7$

$-7x=-7$

$\boxed{x=1}$

$5(1)+2y+10(0)=7$

$5+2y=7$

$\boxed{y=1}$

(14) $\begin{cases} x+z=6 & a \\ 2x-4y+3z=17 & b \\ -3x+y-2z=-13 & c \end{cases}$

$\begin{cases} 2x-4y+3z=17 & b \\ -12x+4y-8z=-52 & c \end{cases}$

$-10x-5z=-35$

$2x+z=7$

$\begin{cases} x+z=6 \\ 2x+z=7 \end{cases}$

$-x=-1$

$\boxed{x=1}$

$2(1)+z=7$

$2+z=7$

$\boxed{z=5}$

$-3(1)+y-2(5)=-13$

$-3+y-10=-13$

$\boxed{y=0}$

15. $\begin{cases} a: & 2x - y + z = 2 \\ b: & -3x + y - z = -3 \\ c: & -2x + y - z = -2 \end{cases}$

$a+b \rightarrow -x = -1$

$x = 1$

$a+c \rightarrow 0 = 0$

$\boxed{\text{Infinite Solutions}}$

$a \rightarrow 2(1) - y + z = 2$

$y = z$

$b \rightarrow -3(1) + y - z = -3$

$y = z$

$c \rightarrow -2(1) + y - z = -2$

$y = z$

16. $\begin{cases} a: & 5x - y + 3z = 8 \\ b: & 5x + 2y - 4z = -32 \\ c: & -5x - 3y + z = 8 \end{cases}$

$b+c \rightarrow -y - 3z = -24$

$a+c \rightarrow -4y + 4z = 16$

$y - z = -4$

$\begin{cases} -y - 3z = -24 \\ y - z = -4 \end{cases}$

$-4z = -28$

$\boxed{z = 7}$

$y - 7 = -4$

$\boxed{y = 3}$

$-5x - 3(3) + 7 = 8$

$-5x - 9 + 7 = 8$

$-5x - 2 = 8$

$-5x = 10$

$\boxed{x = -2}$

(17)
$$\begin{cases} a: & -2x+y-8z=-9 \\ b: & 3x+2y+z=6 \\ c: & 5x-y-7z=-3 \end{cases}$$

$a+c \rightarrow 3x-15z=-12$

$x-5z=-4$

$$\begin{cases} b: & 3x+2y+z=6 \\ 2c: & 10x-2y-14z=-6 \end{cases}$$

$13x-13z=0$

$x-z=0$

$x=z$

$z-5z=-4$

$-4z=-4$

$\boxed{z=1}$

$\boxed{x=1}$

$3(1)+2y+1=6$

$3+2y+1=6$

$\boxed{y=1}$

(18)
$$\begin{cases} a: & 5x+2y+2z=-2 \\ b: & 6x+9y-4z=4 \\ c: & 10x-4y+3z=-3 \end{cases}$$

$$\begin{cases} 9a: & 45x+18y+18z=-18 \\ 2b: & 12x+18y-8z=8 \end{cases}$$

$33x+26z=-26$

$$\begin{cases} 2a: & 10x+4y+4z=-4 \\ c: & 10x-4y+3z=-3 \end{cases}$$

$20x+7z=-7$

$$\begin{cases} 33x+26z=-26 \\ 20x+7z=-7 \end{cases}$$

$$\begin{cases} 231x+182z=-182 \\ 520x+182z=-182 \end{cases}$$

$-289x=0$

$\boxed{x=0}$

$20(0)+7z=-7$

$\boxed{z=-1}$

$5(0)+2y+2(-1)=-2$

$2y-2=-2$

$\boxed{y=0}$

(19) $\begin{cases} a & -9x+y+8z=-10 \\ b & 3x+4y-2z=-11 \\ c & 2y-5z=4 \end{cases}$

$\begin{cases} a & -9x+y+8z=-10 \\ 3b & 9x+12y-6z=-33 \end{cases}$

$13y+2z=-43$

$\begin{cases} 2y-5z=4 \\ 13y+2z=-43 \end{cases}$

$\begin{cases} 4y-10z=8 \\ 65y+10z=-215 \end{cases}$

$69y=-207$

$\boxed{y=-3}$

$13(-3)+2z=-43$

$-39+2z=-43$

$2z=-4$

$\boxed{z=-2}$

$3x+4(-3)-2(-2)=-11$

$3x-12+4=-11$

$3x-8=-11$

$3x=-3$

$\boxed{x=-1}$

(20) $\begin{cases} a & x+y-z=2 \\ b & -x-y+z=3 \\ c & x+y+z=6 \end{cases}$

$a+b \rightarrow 0=5$

$\boxed{\text{No solutions}}$